The step estimator and the at-cost instrument

Spanish electricity market project

27 October 2026

This note writes out what `step_engine.py`, `leg_b_levels.py` and `price_taking_simulation.py` compute, and maps each symbol to the data. The scripts are the authority: where this note and the code disagree, the code is right.

1 A worked example

A firm offers a hydro step at €62 per MWh, and its water value is €55, so the markup is €7. In 4 of the 40 simulated hours this step sets the price:

draw	firm's net position n (MW)	rivals' slope s (MW per €)
1	300	100
2	200	50
3	100	50
4	200	200

The pooled incentive is $\tilde{X} = (300 + 200 + 100 + 200)/(100 + 50 + 50 + 200) = 800/400 = €2$. A price-taker would offer at €55 (markup 0); a firm exercising full market power would offer at €57 (markup $\tilde{X}$). The parameter β is the share of $\tilde{X}$ that appears in markups, across many steps.

2 The model

Setting. Firm f submits a step curve for hour H before it knows its residual demand. That uncertainty is represented by $R = 40$ equally likely draws $r = 1, \ldots, R$ of the rivals' curve (Section 7). Let $RD_r(p)$ be f's residual demand in draw r: market demand minus rivals' supply at price p. At the clearing price p_r^*, f's net position (sales minus purchases, exposed volume only) is

$$n_r = RD_r(p_r^*),$$

and the slope of its residual demand is

$$s_r = |RD_r'(p_r^*)| \approx \frac{\text{rivals' MW offered within } \pm h \text{ of } p_r^*}{2h}, \qquad h = €5.$$

The draws. Take H = 20:00 on Tuesday 14 March 2023. The firm bids at noon on 13 March without knowing how rivals will bid. The candidates are the 20:00 hour of each day from 14 February to 11 April, excluding 14 March itself: 56 evenings. Of these, the 40 whose forecast of net load (REE's D−1 demand forecast minus its wind and solar forecasts) is nearest 14 March's are kept. In each kept evening the market's real curves are taken, f's own offers removed, and the rest shifted sideways by the difference in forecasts:

$$\Delta = (1 - d_f)\,\Delta F_{\text{demand}} - (1 - w_f)\,\Delta F_{\text{wind}} - (1 - s_f)\,\Delta F_{\text{solar}},$$

with d_f, w_f, s_f the firm's shares of buy offers, wind and solar. f's actual curve for 14 March is then cleared against each of the 40 rival curves. The idea is that how rivals bid on similar evenings stands in for f's uncertainty about how they will bid on this one. Its weakness: the spread across evenings mixes true uncertainty with real changes between days (a move in the gas price from 9 to 14 March), so it may overstate that uncertainty.

One step. Step k of unit u is offered at price q_k; the unit's marginal (opportunity) cost is c_k. Let

$$M_k = \{\, r : p_r^* = q_k \,\}$$

be the draws in which step k sets the price. In the other draws a small change in q_k does not move the price, so it does not move profit.

First-order condition. In a draw $r \in M_k$, profit as a function of the price is $\pi_r(p) = (p - c_k)\, RD_r(p)$ near $p = q_k$, plus terms that do not depend on p. So

$$\pi_r'(q_k) = RD_r(q_k) + (q_k - c_k)\, RD_r'(q_k) = n_r - (q_k - c_k)\, s_r.$$

Raising the price by one euro earns one euro on each of the n_r net MW and loses the margin $q_k - c_k$ on the s_r MW that rivals take. The step's price is optimal in expectation when

$$\sum_{r \in M_k} \big[\, n_r - (q_k - c_k)\, s_r \,\big] = 0 \quad \Longleftrightarrow \quad q_k - c_k = \frac{\sum_{r \in M_k} n_r}{\sum_{r \in M_k} s_r} \equiv \tilde{X}_k.$$

This is the condition of Wolak (2007), Kastl (2011) and Reguant (2014). $\tilde{X}_k$ is a ratio of sums, not a mean of the ratios n_r / s_r.

Conduct. Nesting price-taking ($\beta = 0$) and full market power ($\beta = 1$):

$$y_k \equiv q_k - c_k = \beta\, \tilde{X}_k + \alpha_{u,m} + \varepsilon_k, \tag{1}$$

where $\alpha_{u,m}$ is a unit $\times$ month fixed effect (unit $\times$ day in some rows) and ε_k is everything unobserved, including any error in c_k. Observations are steps that set the price in at least one draw ($|M_k| > 0$), weighted by $|M_k|$, with a mean slope $\sum_{M_k} s_r / |M_k|$ of at least 20 MW per €. For hydro, $c_k = \lambda_{30s}^*$, the water value, and y_k is `gap30`; with unit $\times$ day effects the daily cost is absorbed and y_k is the offer price.

3 Why $\tilde{X}$ needs an instrument

$\tilde{X}_k$ is built on the firm's own curve. Both M_k (in which draws step k sets the price) and n_r (the firm's position there) depend on the firm's own offer prices, and those contain ε. A step priced high because its true cost is high sets the price in different, higher-priced draws than a step priced low. So $\mathrm{Cov}(\tilde{X}_k, \varepsilon_k) \neq 0$, and least squares on (1) is biased.

4 The at-cost instrument Z^b

Counterfactual curve. Move every one of f's storage and pumped-storage hydro steps priced above €1 to its unit's water value, $q_j \to c_j$, and leave every other offer as submitted, including f's own non-hydro offers. Re-clear each draw against the same rivals' curves, giving $\tilde{p}_r$, $\tilde{n}_r$ and $\tilde{s}_r$.

Cost point and near draws. Let $\bar{c}_d$ be the median water value across hydro units on day d, and N_k the 5 draws whose $\tilde{p}_r$ is nearest $\bar{c}_d$. Then

$$Z_k^b = \frac{\sum_{r \in N_k} \tilde{n}_r}{\sum_{r \in N_k} \tilde{s}_r},$$

defined when $\sum_{N_k} \tilde{s}_r/5$ is at least 20 MW per €. It is $\tilde{X}$ evaluated where the firm is a price-taker in its hydro, that is, at $\varepsilon = 0$: the approximation to optimal instruments of Berry, Levinsohn and Pakes (1999). Using the class median $\bar{c}_d$ rather than the unit's own c_k keeps an error in one unit's cost out of the instrument; Z^c uses c_k itself.

Assumption (Exclusion)**.** $\mathbb{E}[\varepsilon_k \mid Z_k^b, \alpha_{u,m}] = 0$.

Z^b contains neither q_k nor c_k. It moves with rivals' curves, the forecasts that select and shift the draws, f's non-hydro offers and $\bar{c}_d$. Relevance: $\text{corr}(\tilde{X}, Z^b) = 0.44$, first-stage F about 90 to 410 across specifications.

What would violate it.

1. The unobserved cost moves with the shape of rivals' curves: a firm values its water more on days when rivals' supply near the price is thin. The planted test (`-sim-scale`: price-takers at $m_f \lambda^*$, with m_f each firm's ratio of offer to λ^*) is one check; it moves β by -0.01 to -0.05.

2. f's own non-hydro offers are kept as submitted, so Z^b is not wholly free of f's choices.

A wrong cost level does not bias β. If $c_k = c_k^{\text{true}} + \delta_{u,m}$, the error is constant within a unit-month and is absorbed by $\alpha_{u,m}$. For the same reason the test cannot confirm the cost level. That rests on separate checks: EDP's revealed water value is within 6% of λ^*.

5 The selection control

A step enters the sample only if some draw's price reaches it, so the kept steps' markups rise with how far the hour's prices sit above the unit's cost, and the instruments move with that too. Let $\bar{\tilde{p}}_k$ and σ_k be the mean and standard deviation of the draws' prices on the at-cost curve, and

$$P_k^{\text{rel}} = \bar{\tilde{p}}_k - c_k.$$

The corrected specification gives each unit i its own slope:

$$y_k = \beta \, \tilde{X}_k + \gamma_i \, P_k^{\text{rel}} + \alpha_{u,m} + \varepsilon_k,$$

with $\tilde{X}_k$ instrumented by Z_k^b. The variants add $\left(P_k^{\text{rel}}\right)^2/100$ or σ_k, each also with a slope per unit; `none` drops the control. The draws' prices are taken on the at-cost curve, so the firm's own offers do not enter the control either.

6 The null band

Simulated price-takers. For seed s, replace f's hydro steps by $c_j + \eta_j^{(s)}$, where $\eta^{(s)}$ is the unit's own markups reshuffled, which ignores f's stake in the price. Then $\beta = 0$ by construction. Rebuild $\tilde{X}$, Z^b and the sample on the same draws and estimate $\hat{\beta}^{(s)}$.

Reading rule (fixed 2026-10-08, before the actual offers were read). With the `level` control,

$$B = \Big[\min_{s\geq 1} \hat{\beta}^{(s)},\ \max_{s\geq 1} \hat{\beta}^{(s)} \Big].$$

The actual offers ($s = 0$) differ from price-takers only if their Anderson–Rubin 95% set does not meet B. Full market power is rejected if 1 lies outside that set. More seeds can only widen B, so a "differs" on two seeds is provisional; seeds 3 to 10 are the next run.

7 Where each symbol comes from

symbol	data	file or code
q_k, f's curve	OMIE day-ahead bid curves (`curva_pbc_uof`); the effective curve: accepted rows below p^*, offered rows at and above it	`step_curves.py` → `own_`, `market_YYYYMM`
f: whose units	OMIE unit listings with ownership shares, dated	`lib/unit_ownership.py`, `owner_groups.parquet`
draws r	40 hours at the same clock hour within ±28 days, nearest in REE's D−1 forecast of net load (ESIOS 1775, 1777, 1779), shifted to H's forecasts	`step_engine.py`
n_r	f's exposed sales minus purchases at p^*_r; regulated pass-through suppliers excluded	`lib/omie_owner.py`
s_r	rivals' MW within ±€5 of p^*_r, over €10; undefined below 5 MW per €	`step_engine.py`
c_k, hydro	λ^*_{30s}: OMIP forwards × last year's hourly shape, the unit's running share, MITECO or SNIRH reservoir fill; known at D−2	`hydro_cost_daily.parquet`
c_k, thermal	MIBGAS gas and EUA carbon (coal: API2 over the ECB rate) divided by JRC efficiency; prices final at noon on D−1	`implied_marginal_cost_daily.parquet`, `_bid` columns
$\sum_{M_k} n_r$, $\sum_{M_k} s_r$, $\lvert M_k\rvert$		`hmoments_YYYYMM`: `m_sn`, `m_s`, `m_n`
$\sum_{N_k} \tilde{n}_r$, $\sum_{N_k} \tilde{s}_r$		`hcompcost_YYYYMM`: `kb_sn`, `kb_s`
$\tilde{\bar{p}}_k$, σ_k		`hcompcostprice_YYYYMM`: `cp_mean`, `cp_sd`
$\hat{\beta}^{(s)}$ inputs	simulated price-takers, seeds 1–3	`hsim`s`_moments`, `hsim`s`_units`
planted test	price-takers at $m_f \lambda^*$	`hsimw1_*`

Thermal uses the same caches without the `h` prefix, with f's gas and coal steps in place of its hydro steps.

What the October rebuild changes. Only the membership of f. ADOURO's Douro dams are Movhera's, not EDP's, from 2020-12-17 to 2021-09-08, and Castejón's represented units are TotalEnergies', not EDP's, from February 2024 to January 2025. In those months the own curves of EDP, Movhera and TotalEnergies change, and with them n_r, M_k, the rivals' curves, $\tilde{X}$, Z^b and P^{rel}. Every other firm's rows were checked to be identical. The costs and the formulas above are unchanged.